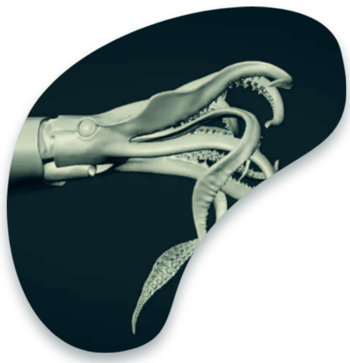

Update of the SPiCT applied to *Dosidicus gigas* in FAO Area 87

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Category: Working paper summary

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1 Document

A stock assessment paper by Chile (Payá 2025), presented as SC13-SQ06 to the 13th Meeting of the SPRFMO Scientific Committee (8–13 September 2025, Wellington, New Zealand). Dated September 2025.

2 Subject

An updated stock assessment of jumbo squid (*Dosidicus gigas*) in FAO Area 87 — which spans the SPRFMO high-seas area plus the Ecuadorian, Peruvian and Chilean EEZs. Its main advance over previous assessments is the use of data up to 2024, reducing the two-year delay of earlier work and bringing the assessment closer to in-season conditions.

3 Methods

The assessment applies the **Stochastic Production model in Continuous Time (SPiCT)** — a state-space, Pella-Tomlinson surplus production model that separates random stock-dynamics variability from observation error and allows assessment at monthly/quarterly steps. Six country CPUE abundance indices were compiled (two Chinese fleet indices, Chinese Taipei, Korea, Chile, Peru), with an updated Chinese index provided by Chinese scientists. A **sensitivity analysis of seven cases** tested the effect of using a single global index versus six country-specific indices, fixing the Schaefer production curve, and applying a prior on the intrinsic growth rate r. Two data spans were compared: full series (2001–2024) and short series (2012–2024).

4 Key results

- Total squid catch fell sharply — a **52% reduction in 2024** — and the abundance indices show a **general decreasing trend**.
- Biomass rose from 2001 to ~2010, then declined; fishing mortality (F) shows the inverse pattern, rising steeply in recent years.
- r and K varied widely across cases and were negatively correlated, yielding an **MSY of ~950 kt** (range 906–950 kt across cases). r estimates (~0.8–2.4) sit in the range expected for a fast-growing, short-lived squid when priors are used.
- **2025 stock status is uncertain**: of the seven cases, four fell in the Kobe red zone, one yellow, two green. Cases fitted to the six country indices placed the stock in the red zone.

- Case 7 (free productivity curve, six indices, r prior) had the **best retrospective pattern** and estimated a 2025 status in the red zone with a 2026 TAC of **467 kt** (similar to the 2024 catch).
- 2026 TAC under F_{MSY} ranged from **267 kt (Case 6) to 989 kt (Case 1)**.
- The short-series (2012–2024) sensitivity gave similar MSY (941–950 kt) and TAC (271–756 kt); two of seven cases did not converge, and most placed 2025 in the red zone.

5 Main conclusion

Stock status is highly uncertain but the most reliable case (best retrospective pattern) places the 2025 stock in the Kobe **red zone** — i.e. potentially overfished and/or undergoing overfishing — a notable contrast with the "healthy" 2022 status discussed in 2024, driven by the 52% catch drop and declining indices. MSY converges around ~950 kt, but reference points and TAC advice remain sensitive to the choice of abundance index.

6 Way forward

Improve and standardize the CPUE abundance indices (incorporating spatiotemporal variability and different squid sizes across regions), since SPiCT assumes indices are unbiased indicators of the whole stock. Account for the **three phenotypes** of *D. gigas* (which likely have distinct growth rates and carrying capacities) and for environmentally driven interannual variability in productivity, which the single-stock process-error assumption may not fully capture.

References

Payá, Ignacio S. 2025. *Update of the Stochastic Production Model in Continuous Time (SPiCT) Applied to Dosidicus gigas in the FAO Area 87.* Scientific Committee Document Nos. SC13-SQ06. Instituto de Fomento Pesquero (IFOP), Stock Assessment Department.